

Week 3 - Day 5

Decision Trees & Overfitting

5-Minute Review Notes + Practical Results

Day Goal

Understand how Decision Trees make predictions, how Gini impurity guides splits, how tree depth controls complexity, and how to recognize underfitting and overfitting.

Day 5 Status: COMPLETE

1. Decision Tree Basics

A Decision Tree makes a prediction by asking a sequence of feature-based questions. Each answer follows a branch until the observation reaches a leaf containing the final prediction.

Root node -> decision/question -> branch -> leaf -> prediction

Part	Meaning
Root node	First/top decision
Internal node	Another decision inside the tree
Branch	Path after a decision
Leaf node	Final prediction

2. Gini Impurity

Gini impurity measures how mixed the classes are inside a node. Trees prefer splits that create purer groups.

Lower Gini -> purer node

Higher Gini -> more mixed node

10 Churn, 0 Stay -> Gini = 0 -> completely pure

5 Churn, 5 Stay -> highly mixed

3. Scikit-Learn Workflow

```
model = DecisionTreeClassifier(random_state=42)
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
```

4. Overfitting vs Underfitting

Pattern	Likely Interpretation
Very high train, much lower test	Possible overfitting
Low train, low test	Possible underfitting
Strong train and strong test	Better evidence of generalization

Overfitting: the model learns training-specific details/noise and performs noticeably worse on unseen data.

Underfitting: the model is too simple to learn the important pattern.

Important: an unrestricted tree does **not automatically** mean overfitting. We judge overfitting from generalization performance.

5. max_depth

`max\_depth` limits how deep the tree can grow and therefore controls model complexity.

```
max_depth=1 -> very simple
max_depth=3 -> moderate
max_depth=None -> unrestricted
```

A useful principle: if multiple models perform equally well on unseen data, prefer the simpler model unless there is a good reason not to.

6. Today's Dataset

Features: Monthly_Charges, Contract_Months, Support_Calls

Target: Churn (0 = Stay, 1 = Churn)

```
Stay = 25 / 50 = 50%
Churn = 25 / 50 = 50%
-> perfectly balanced dataset
```

7. Unrestricted Tree Result

```
Training Accuracy = 1.00
Testing Accuracy = 1.00
```

There is **no train/test evidence of overfitting** on this split because both scores are 100%. However, the dataset is small and simple, so this does not prove perfect real-world generalization.

8. Tree Depth Experiment

Depth	Train Accuracy	Test Accuracy
1	0.918919	1.000000
2	0.918919	1.000000
3	0.945946	1.000000
4	0.972973	1.000000
5	1.000000	1.000000
None	1.000000	1.000000

Selected depth: 1

Depth 1 achieved the same 100% test accuracy as all deeper trees while remaining the simplest model. There was no test-performance benefit from extra complexity.

9. Feature Importance

For the selected depth-1 model:

Feature	Importance
Support_Calls	1.0
Monthly_Charges	0.0
Contract_Months	0.0

Interpretation: this particular depth-1 tree used only **Support_Calls** for its split. This does not prove the other features are useless.

Feature importance does not mean causation.

10. Correct Feature-Importance Code

```
importance_df = pd.DataFrame({
    "Feature": X.columns,
    "Importance": best_model.feature_importances_
})

importance_df = importance_df.sort_values(
    by="Importance", ascending=False
)
```

11. New Customer Predictions

Monthly Charges	Contract Months	Support Calls	Prediction
40	24	0	Stay (0)
85	6	3	Churn (1)
115	1	6	Churn (1)

12. Day 5 Assessment - Correct Answers

Q	Answer
1	A Decision Tree asks feature-based questions and follows branches until it reaches a leaf prediction.
2	The root is the first/top decision; a leaf contains the final prediction.
3	Gini measures how mixed classes are. Lower Gini = purer; higher Gini = more mixed.
4	Node A (10 Churn, 0 Stay) is purer because it contains only one class.
5	Train 1.00 vs Test 0.68 suggests possible overfitting because the generalization gap is large.
6	Train 0.61 vs Test 0.59 suggests underfitting because both scores are poor.
7	max_depth controls how deep/complex the tree can grow.
8	Choose depth 3 because it has strong test performance and a much smaller train/test gap than depth 10.
9	No. Feature importance is not proof of causation.
10	Highest training accuracy can come from memorizing training data rather than generalizing.

13. Key Corrections From Today's Work

Correction	Remember
Overfitting	Train=1.0 and Test=1.0 is not evidence of overfitting by itself.
Gini	Pure node = low Gini, not high Gini.
Feature importance	Use best_model.feature_importances_ when interpreting your selected tree.
Prediction goal	The model predicts whether a customer churns; it does not directly explain why they churn.

14. Quick Review Cheat Sheet

Decision Tree -> learned feature questions
Root -> first decision
Leaf -> final prediction

Lower Gini -> purer node
Higher Gini -> more mixed node

High train + much lower test -> overfitting risk
Low train + low test -> underfitting risk

max_depth -> controls complexity
Equal test performance -> prefer simpler model
Feature importance != causation

15. Mental Model

Too simple
-> Underfitting

Appropriate complexity
-> Better generalization

Too complex
-> Can memorize noise
-> Overfitting risk

16. Completion Summary

You built an unrestricted Decision Tree, evaluated training and testing performance, compared multiple tree depths, selected a simpler depth-1 model, interpreted feature importance, and predicted churn for new customers.

The most important lesson is that good Machine Learning is not about maximizing training accuracy. It is about learning patterns that generalize to unseen data.

Week 3 Progress: 5/7 complete

Completed: Day 5 - Decision Trees & Overfitting

Next: Week 3 - Day 6